

ServiceGuard AI

Pre-Implementation Documentation Pack

AI-Powered SaaS Service Operations & SLA Risk Intelligence Platform

Purpose

This pack defines the product, business problem, domain model, requirements, architecture, APIs, data, ML, RAG, agent, security, testing, deployment and implementation plan before coding begins.

Recommended document order

Order	Document	Decision it establishes
01	Product Definition & Vision	What the product is and why it exists
02	PRD / Functional Requirements	What users must be able to do
03	Business & Use-Case Specification	Who uses it and operational workflows
04	Dataset & Data Dictionary	What data exists and how it is generated
05	HLD / System Architecture	How services interact
06	LLD / Backend Design	How Java modules/classes behave
07	Database Design	Tables, relationships, indexes and views
08	API Contract	Stable Java/Python/frontend contracts
09	ML Design	Risk prediction problem and lifecycle
10	RAG & Knowledge Architecture	Runbook/SOP ingestion and retrieval
11	AI Agent Design	Tools, state, guardrails and reasoning workflow
12	Security & Threat Model	Trust boundaries and controls
13	Testing & Quality Strategy	How correctness will be proven
14	Deployment & Operations	How the system runs and is monitored
15	Implementation Roadmap	Build sequence and definition of done

Architecture principle

Java owns the application; Python owns intelligence. PostgreSQL is the operational source of truth. The AI layer is evidence-driven and advisory, with human approval for high-impact actions.